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# Part 1. Using identity.

$(a+b)^2 = a^2 + 2ab + b^2$

1. $(x+4)^2 = (x)^2 + 2(x)(4) + (4)^2$
$= x^2 + 8x + 16$

4. $(2x+3)^2 = (2x)^2 + 2(2x)(3) + (3)^2$
$= 4x^2 + 12x + 9$

7. $(4k+1) = (4k)^2 + 2(4k)(1) + (1)$
$= 16k^2 + 8k + 1$

10. $(\frac{1}{2}x+4)^2 = (\frac{1}{2}\cdot\frac{1}{2}\times x) + 2(\frac{1}{2}x)(4) + (4)^2$
$= \frac{1}{4}x^2 + 4x + 16$

$(a-b)^2 = a^2 - 2ab + b^2$

2. $(y-6)^2 = (y)^2 - 2(y)(6) + (6)^2$
$= y^2 - 12y + 36$

5. $(5m-2)^2 = (5m)^2 - 2(5m)(2) - (2)^2$
$= 25m^2 - 20m - 4$

8. $(7-2y)^2 = (7)^2 - 2(7)(2y) + (2y)^2$
$= 49 - 28y + 4y^2$
$= 4y^2 - 28y + 49$

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# Part 1. Using Identity.

$(a+b)(a-b) = a^2 - b^2$

3. $(a+9)(a-9) = (a)^2 - (9)^2$

$= a^2 - 81$

6. $(3p+4q)(3p-4q) = (3p)^2 - (4q)^2$

$= 9p - 16q$

9. $(10a+3b)(10a-3b) = (10a)^2 - (3b)^2$

$= 100a - 9b$

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# Part 2 . Backward the identity

Factorise the algebraic Expression.

11. $x^2 + 10x + 25$

breakdown

$a = \sqrt{x^2} = x$

$b = \sqrt{25} = 5$

id : $(a+b)^2$

$x^2 + 10x + 25 = (x + 5)^2$

12. $y^2 - 14y + 49$

$a = \sqrt{y^2} = y$

$b = \sqrt{49} = 7$

id : $(a-b)^2$

$y^2 - 14y + 49 = (y - 7)^2$

13. $m^2 - 36$

$a = \sqrt{m^2} = m$

$b = \sqrt{36} = 6$

id = $a^2 - b^2 = (a+b)(a-b)$

$m^2 - 6^2 = (m+6)(m-6)$

14 $4a^2 + 12a + 9$

$a = \sqrt{4a^2} = 2a$

$b = \sqrt{9} = 3$

id : $(a+b)^2$

$4a^2 + 12a + 9 = (2a + 3)^2$

15. $16x^2 - 24x + 9$

$a = \sqrt{16x^2} = 4x$

$b = \sqrt{9} = 3$

id : $(a-b)^2$

$16x^2 - 24x + 9 = (4x - 3)^2$

16. $25p^2 - 64q^2$

$a = \sqrt{25p^2} = 5p$

$b = \sqrt{64q^2} = 8q$

id : $a^2 - b^2 = (a+b)(a-b)$

$25p^2 - 64q^2 = (5p + 8q)(5p - 8q)$

17. $81 - b^2$

$a = \sqrt{81} = 9$

$b = \sqrt{b^2} = b$

id : $a^2 - b^2 = (a+b)(a-b)$

$81 - b^2 = (9 + b)(9 - b)$

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18. $9x^2 + 30xy + 25y^2$

a: $\sqrt{9x^2}$ → $3x$

b: $\sqrt{25y^2}$ → $5y$

id : $(a+b)^2$

$9x^2 + 30xy + 25y^2 = (3x + 5y)^2$

19. $49m^2 - 144n^2$

a: $\sqrt{49m^2}$ → $7m$

b: $\sqrt{144n^2}$ → $12n$

id : $a^2 - b^2 = (a+b)(a-b)$

$49m^2 - 144n^2 = (7m + 12n)(7m - 12n)$

20. $2x^2 - 18 =$

1. Simplify by find the common factors lowest

$2x^2 - 18$

L

2. Find the fachrs

$2x^2 - 18 = 2(x^2 - 9)$

$= 2(x+3)(x-3)$

id : $a^2 - b^2 = (a+b)(a-b)$

$x^2 - 9 = (x+3)(x-3)$

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## Part 3. Solve the problem

21. $41^2 = (40+1)^2$

$(a+b)^2 = a^2 + 2ab + b^2$

$(40+1)^2 = (40)^2 + 2(40)(1) + (1)^2$

$= 1600 + 80 + 1$

$= 1681$

22. $98^2 = (100-2)^2$

$(a-b)^2 = a^2 - 2ab + b^2$

$(100-2)^2 = (100)^2 - 2(100)(2) + (2)^2$

$= 10000 - 400 + 4$

$= 9604$

23. $102(98)$

$102 = 100 + 2$ $\quad 98 = 100 - 2$

$(a+b)(a-b) = a^2 - b^2$

$(100+2)(100-2) = 100^2 - 2^2$

$= 10000 - 4$

$= 9996$

24. $\underbrace{(x+3)}_{a}^2 - \underbrace{(x-3)}_{b}^2 = 24$

(id) $a^2 - b^2 = (a+b)(a-b)$

$a = x+3$

$b = x-3$

$(x+3)(x-3) = 24$

$(a+b)(a-b) = c$

$(x+3+x-3)(x+3-x+3) = 24$

$(2x)(6) = 24$

$12x = 24$

$x = 2$

$(x+3)$
$(x-3)$
$2x$ $+$

$(x+3)$
$-(x-3)$ $\quad -(-x+3)$
$6$

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## Part 3. Solve the problem

(1d) $a+b$, $ab$ ] $(a+b)^2 = a^2 + b^2 + 2ab$

$\therefore$ $a+b = 7$

$ab = 10$

Find value of $a^2+b^2$

$(a+b)^2 = a^2 + b^2 + 2ab$

$7^2 = a^2 + b^2 + 2(10)$

$49 = a^2 + b^2 + 20$

$49 - 20 = a^2 + b^2$

$29 = a^2 + b^2$

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## Part 4. Explain the step

26. Explain the steps to expand $(3x-5)^2$:
Identify which special identity and clearly state what your a and b values before solving.

$a = 3x$ $\quad b = 5$

identity : $(a-b)^2 = a^2 + 2ab + b^2$

$(3x-5)^2 = 3x^2 + 2(3x)(5) + 5^2$

$= 9x^2 + 30x + 25$

27. Identify the error

$(x+7)^2 = x^2 + 49 \boxed{+14x}$

The missing part:

term: $2ab$

$= +14x$

identity $(a+b)^2 = a^2 + 2ab + b^2$

$(x+7)^2 = x^2 + 2(x)(7) + 49$

$= x^2 + 14x + 49$

28. How to recognize the perfect square:

$16x^2 - 40x + 25$

id: $a^2 + 2ab + b^2 = (a-b)^2$

$a = \sqrt{16x^2}$

$= 4x$

First term

$2ab$

$2(4x)(5)$

$= 40x$

middle term

$b = \sqrt{25}$

$= 5$

last term

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29. Write down step by step to factorise $50x^2 - 8$.
Why you can't apply the difference of two squares immedietly ?

identity for the difference of two squares

$$a^2 - b^2 = (a + b)(a - b)$$

[a] To use this both a and b must be **square number**

1. Factorise to [find near square number]

| 50 | |
|---|---|
| 1 | 50 |
| ~~2~~ | (25) |
| 5 | 10 |

| 8 | |
|---|---|
| 1 | 8 |
| ~~2~~ | (4) |

square number with factor of 2

ii Apply the identity

$$2(25x^2 - 4^2) = 2(5x + 2)(5x - 2)$$

$a = \sqrt{25x^2} = 5x$

$b = \sqrt{4^2} = 2$

30. Explain how identity $(a+b)(a-b) = a^2 - b^2$
quickly calculate
"directly calculate"

$$85^2 - 15^2 = (85 + 15)(85 - 15)$$
$$= (100)(70)$$
$$= 7000$$

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